c. Electricity

Electricity is essential for our living. We use electricity for cooking, lighting, grinding, watching television etc. Do you know electricity can be used to carry out chemical reactions also? Many chemical reactions which take place with the help of electricity are industrially very important. As you know, water is made of hydrogen and oxygen molecules. When electricity is passed through water which contains small amounts of sulphuric acid, hydrogen and oxygen gases are liberated. Similarly, a concentrated solution of sodium chloride called brine is electrolysed to produce chlorine and hydrogen gases along with sodium hydroxide. This is an important reaction to produce chlorine industrially.

Thus, we can conclude that some chemical reactions proceed only by the passage of electricity. Hence, such reactions are called as **electrochemical reactions** or **electrolysis**.

DO YOU KNOW?

The term electrolysis was introduced by Michael Faraday in the 19th century. The word electrolysis is a combination of two terms 'electron' and 'lysis'. Electron is related to electricity and lysis means decomposition.



d. Heat

Food is important for our survival and also for the survival of many other living beings. Have you ever closely watched your mother cooking food? She boils rice, cooks vegetables, and prepares gravy by heating them over stove. When enough heat is given some chemical reactions take place to convert the raw food (uncooked) items into cooked ones.

You can learn more about this by conducting a reaction in your laboratory.

More to Know

Chemical reactions accompanying evolution of heat are called exothermic reactions. Whereas chemical reactions which involve absorbtion of heat are called endothermic reactions.

Take a small amount of lead nitrate in a dry test tube and heat it gently over a flame. Observe the changes closely. You will hear cracking sound and an evolution of reddish brown coloured gas (nitrogen dioxide). In industries limestone rocks are heated to get quicklime (calcium oxide). Hence, some chemical reactions can be achieved by the supply of heat only. These reactions are called **thermo chemical reactions** or **thermolysis**.

DO YOU KNOW?

Limestone is the raw material for quicklime, slaked lime and cement.



e. Light

What will happen if there is no sunlight? All the living organisms will be affected and there will be no food for us to survive, is n't it? Sunlight is important not only for us but also for plants. As you know photosynthesis ('photo' means light and 'synthesis' means production) is a process in which light energy from the sun is used by the plants to prepare starch from carbon dioxide and water. The sunlight induces the chemical reaction between carbondioxide and water, which finally ends up in the production of starch. Thus, chemical reactions induced by light are called as photochemical reactions.

DO YOU KNOW?

The ultraviolet rays from the sun break ozone (O_3) molecules in the stratosphere into molecular oxygen and atomic oxygen. This atomic oxygen again combines with molecular oxygen to form Ozone.

8th_Science_Unit-10_EM.indd 106 12/1/2022 3:43:21 PM